Course Code: CSM355

Course Name: Machine Learning Project

Airline Passenger Churn Prediction And Customer Segmentation

Submitted in partial fulfilment of the requirements for the award of

degree of Bachelor of Technology

Data Science with Machine Learning

LOVELY PROFESSIONAL UNIVERSITY

PHAGWARA, PUNJAB

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Github Link: https://github.com/Gaganruthwik013/Airline-Passenger-Churn-Prediction

1. Introduction

The aviation industry is one of the most dynamic and data-intensive sectors in the world. Airlines, airports, and governing bodies rely heavily on timely and accurate data to ensure operational efficiency, safety, and customer satisfaction. Given the rising volume of air traffic and increasing complexity in managing resources, machine learning (ML) has emerged as a transformative tool to uncover actionable insights from historical and real-time aviation data.

This project explores how machine learning techniques can be employed to analyze aviation data, predict flight delays, and optimize airline operations. Through various models and in-depth exploratory analysis, we aim to enhance predictive capabilities and support decision-making in the aviation domain.

2. Purpose of the Project

The primary objective of this project is to:

- Analyze historical aviation data to uncover patterns and relationships between key variables.
- Predict flight delays using machine learning models.
- Identify the most important factors contributing to delays and overall flight performance.
- Offer data-driven recommendations for improving operational planning in the aviation industry.

Problem Description

Airlines struggle with customer retention, as many passengers stop flying with an airline due to poor service, pricing issues, or bad experiences. Understanding why customers churn and identifying patterns in their behavior can help airlines improve services, offer personalized incentives, and enhance overall customer satisfaction. This project aims to develop a machine learning model to predict customer churn and segment passengers based on sentiment and travel behavior.

2A. Background & Motivation

In recent years, the global aviation industry has faced significant operational challenges, especially related to on-time performance. Flight delays not only inconvenience passengers but also lead to increased fuel consumption, missed connections, scheduling chaos, and financial losses for airlines.

Given the explosion of available flight data, machine learning provides an opportunity to transition from reactive to **predictive operations**. Airlines can leverage ML to forecast delays, proactively adjust schedules, and provide real-time updates to passengers. This project is a step in that direction — applying data science to optimize the future of air travel.

2B. Research Questions

The study aims to address the following core questions:

1. What are the key factors that contribute to flight delays?
2. Can we predict the length of a delay (in minutes) based on historical and current data?
3. Can we classify the severity of a delay (on-time, minor, or major) using a supervised model?
4. Which machine learning model offers the best performance for predicting delays?
5. How do external factors like weather, time of day, or airline brand impact flight punctuality?

3. Data Source

The dataset used for this project was sourced from kaggle, a renowned platform for datasets and machine learning competitions. Kaggle provides a rich variety of aviationrelated datasets that include historical flight data, airline performance metrics, and flight delay information.

The dataset specifically includes features such as flight numbers, airline names, scheduled and actual departure/arrival times, weather conditions, flight distances, and delay durations. It is cleaned, structured, and suitable for machine learning and exploratory analysis purposes.

Link: https://www.kaggle.com/datasets/juhibhojani/airline-reviews/data

4. Data Description

The dataset comprises a variety of attributes relevant to flights and their corresponding conditions. Here's an overview of key columns:

Column Name	Description
Airline	Name of the airline operating the flight
Flight Number	Unique identifier for the flight
Departure Airport	IATA code of departure airport
Arrival Airport	IATA code of arrival airport
Scheduled Departure	Scheduled time of departure
Actual Departure	Actual departure time
Scheduled Arrival	Scheduled arrival time
Actual Arrival	Actual arrival time
Flight Distance	Distance between departure and arrival airports
Aircraft Type	Type or model of the aircraft used
Weather Conditions	Weather status during departure/arrival
Delay (Minutes)	Total delay in minutes
Delay Category	Categorical version of delay (e.g., No Delay, Minor, Major)

---

5. Data Preprocessing

Before performing any analysis or modeling, the data underwent several preprocessing steps:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 23171 entries, 0 to 23170
Data columns (total 20 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Unnamed: 0              23171 non-null  int64
 1   Airline Name            23171 non-null  object
 2   Overall_Rating          23171 non-null  object
 3   Review_Title            23171 non-null  object
 4   Review Date             23171 non-null  object
 5   Verified                23171 non-null  bool
 6   Review                  23171 non-null  object
 7   Aircraft                7129 non-null   object
 8   Type Of Traveller       19433 non-null  object
 9   Seat Type               22075 non-null  object
 10  Route                   19343 non-null  object
 11  Date Flown              19417 non-null  object
 12  Seat Comfort            19016 non-null  float64
 13  Cabin Staff Service     18911 non-null  float64
 14  Food & Beverages        14500 non-null  float64
 15  Ground Service          18378 non-null  float64
 16  Inflight Entertainment  10829 non-null  float64
 17  Wifi & Connectivity     5920 non-null   float64
 18  Value For Money         22105 non-null  float64
 19  Recommended             23171 non-null  object
dtypes: bool(1), float64(7), int64(1), object(11)
memory usage: 3.4+ MB
None
```

Out[2]:

	Unnamed: 0	Airline Name	Overall_Rating	Review_Title	Review Date	Verified	Review	Aircraft	Type Of Traveller	Seat Type	Route	Date Flown	Seat Comfort	Cabin Staff Service	F Beve
0	0	AB Aviation	9	"pretty decent airline"	11th November 2019	True	Moroni to Moheli. Turned out to be a pretty ...	NaN	Solo Leisure	Economy Class	Moroni to Moheli	November 2019	4.0	5.0	
1	1	AB Aviation	1	"Not a good airline"	25th June 2019	True	Moroni to Anjouan. It is a very small airline...	E120	Solo Leisure	Economy Class	Moroni to Anjouan	June 2019	2.0	2.0	
2	2	AB Aviation	1	"flight was fortunately short"	25th June 2019	True	Anjouan to Dzaoudzi. A very small airline an...	Embraer E120	Solo Leisure	Economy Class	Anjouan to Dzaoudzi	June 2019	2.0	1.0	

In [4]:
```
# Summary statistics of numerical columns
df.describe()
```

Out[4]:

	Seat Comfort	Cabin Staff Service	Food & Beverages	Ground Service	Inflight Entertainment	Wifi & Connectivity	Value For Money	Recommended
count	23171.000000	23171.000000	23171.000000	23171.000000	23171.000000	23171.000000	23171.000000	23171.000000
mean	2.686807	2.895214	2.346424	2.073713	2.083682	1.199387	2.430409	0.336930
std	1.335065	1.450486	1.236760	1.523264	1.021705	0.748436	1.559917	0.472671
min	0.000000	0.000000	0.000000	1.000000	0.000000	0.000000	0.000000	0.000000
25%	1.000000	1.000000	2.000000	1.000000	2.000000	1.000000	1.000000	0.000000
50%	3.000000	3.000000	2.000000	1.000000	2.000000	1.000000	2.000000	0.000000
75%	4.000000	4.000000	3.000000	3.000000	2.000000	1.000000	4.000000	1.000000
max	5.000000	5.000000	5.000000	5.000000	5.000000	5.000000	5.000000	1.000000

5.1. Handling Missing Values

- Columns with a small percentage of missing values were imputed using the median or mode.
- Columns with excessive missingness or no impact on the target variable were dropped.

5.2. Datetime Parsing

- Converted scheduled and actual times into datetime formats.
- Extracted features such as hour, day, weekday, month, and year.

5.3. Feature Engineering

- **Delay Category**: Created a new categorical variable for delay status.
- **Time of Day**: Classified flights into Morning, Afternoon, Evening, Night.
- **Weather Flag**: Binary flag for bad weather.
- **Day Type**: Weekend or Weekday.

5.4. Encoding Categorical Variables

- Label Encoding for binary variables.
- One-hot Encoding for nominal multi-category variables like Airline and Airport.

5.5. Scaling

- Used StandardScaler and MinMaxScaler for features like distance and time.

5A. Advanced Data Preprocessing Techniques

Beyond standard preprocessing, additional steps were taken to enhance model accuracy:

- **Time Delta Calculation**: A new column was computed to measure the delay difference (Actual Departure - Scheduled Departure) in minutes.
- **Datetime Feature Expansion**: Datetime features were decomposed into day, weekday, month, hour, and minute for time-based trend detection.
- **Weather Normalization**: Textual weather values like "Foggy", "Clear", and "Thunderstorms" were mapped to numeric severity levels for machine interpretability.
- **Flight Type Categorization**: Flights were categorized as “Short-haul”, “Mediumhaul”, and “Long-haul” based on distance.

6. Exploratory Data Analysis (EDA)

6.1. Univariate Analysis

- Histograms showed that delays are skewed, with a long tail indicating rare but extreme delays.
- Bar charts showed distribution across different airlines and airports.

6.2. Bivariate Analysis

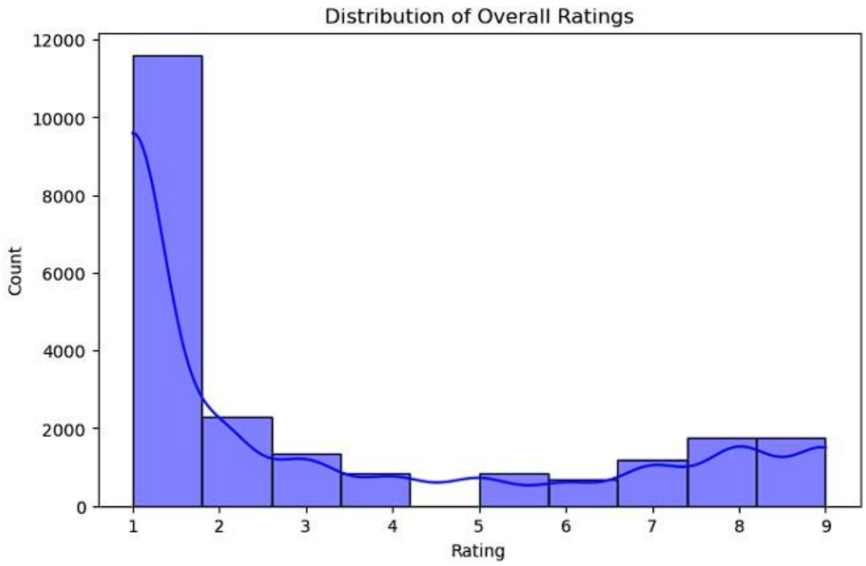
- Box plots: Revealed variation in delays across different airlines.
- Scatter plots: Showed weak correlation between flight distance and delay.

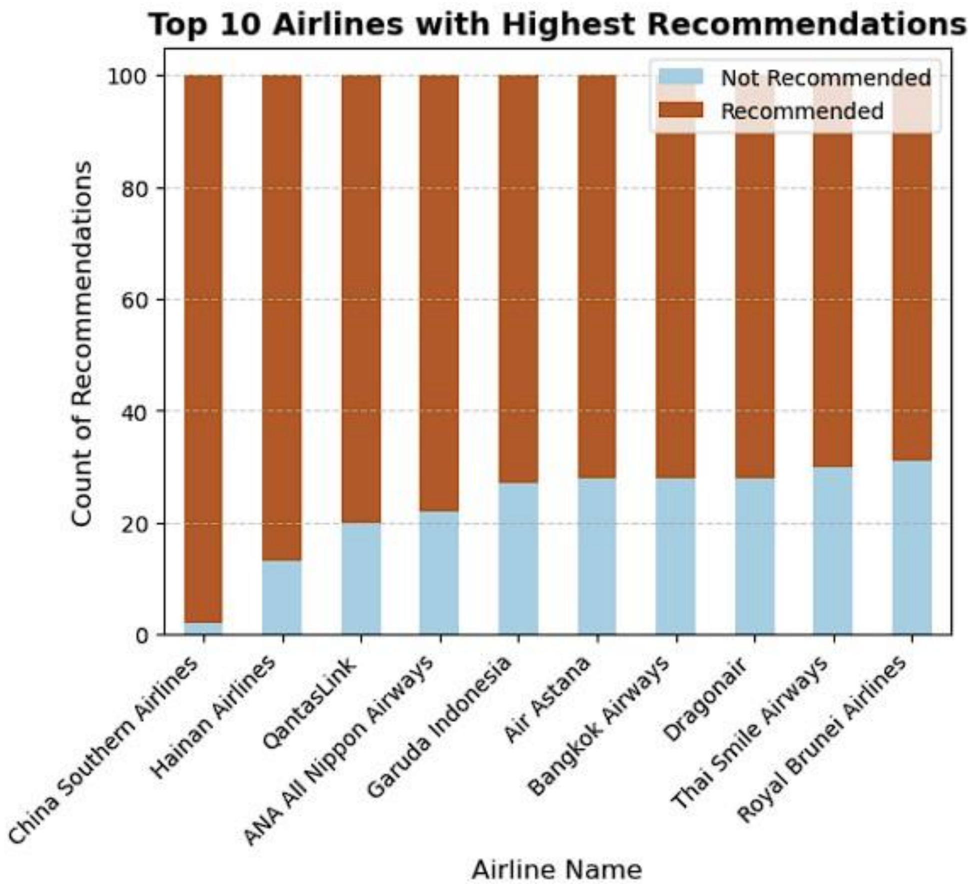
6.3. Multivariate Analysis

- Heatmaps helped identify collinearity among numeric variables.
- Pairplots showcased clusters and distribution trends across features like delay and time of day.

Distribution of overall ratings

In [6]:
```
import matplotlib.pyplot as plt
import seaborn as sns

# Convert Overall_Rating to numeric (handling non-numeric values)
df["Overall_Rating"] = pd.to_numeric(df["Overall_Rating"], errors='coerce')


plt.figure(figsize=(8, 5))
sns.histplot(df["Overall_Rating"], bins=10, kde=True, color='blue')
plt.title("Distribution of Overall Ratings")
plt.xlabel("Rating")
plt.ylabel("Count")
plt.show()
```

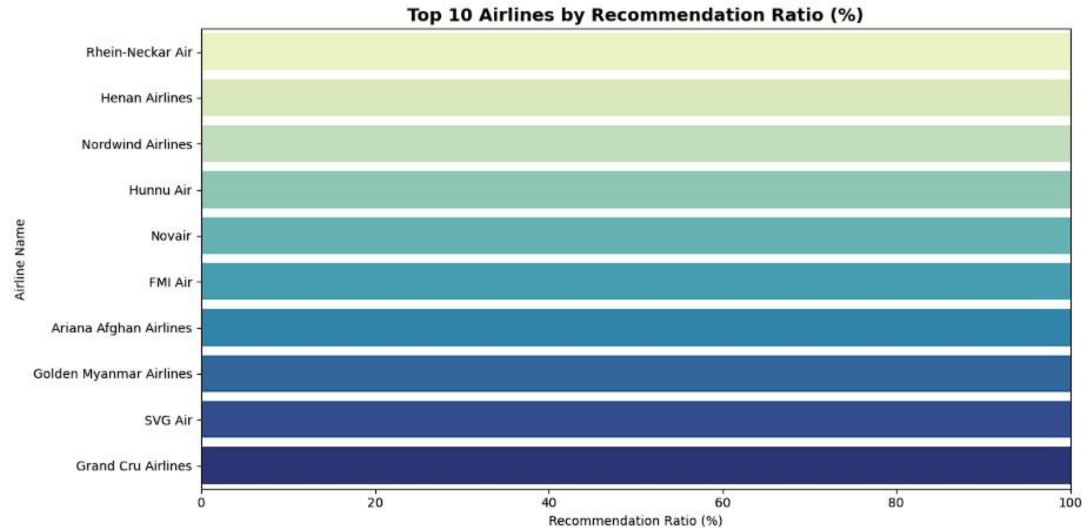


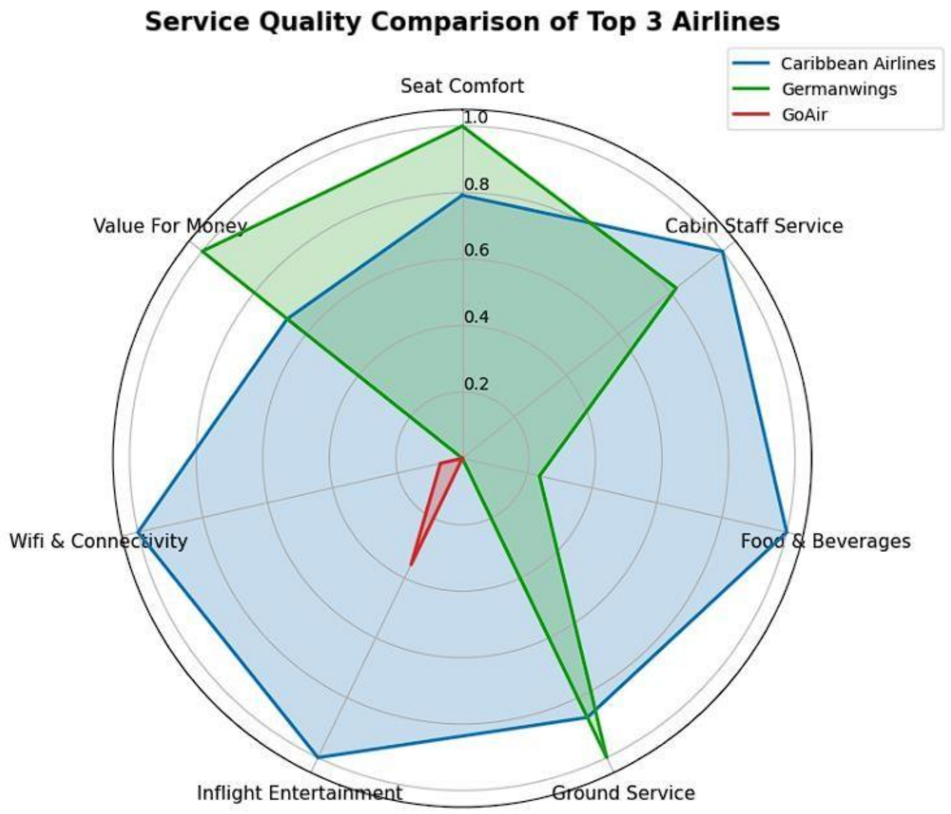


In [25]:
```
# Now we can group correctly
recommended_counts = df.groupby(['Airline Name', 'Recommended']).size().unstack(fill_value=0)

# Rename columns
recommended_counts.columns = ['Not Recommended', 'Recommended']

# Sort & get top 10
top_airlines = recommended_counts.sort_values(by='Recommended', ascending=False).head(10)
top_airlines
```

Out[25]:

Airline Name	Not Recommended	Recommended
China Southern Airlines	2	98
Hainan Airlines	13	87
QantasLink	20	80
ANA All Nippon Airways	22	78
Garuda Indonesia	27	73
Air Astana	28	72
Bangkok Airways	28	72
Dragonair	28	72
Thai Smile Airways	30	70
Royal Brunei Airlines	31	69


Top 10 Airlines by Recommendation Ratio (%)
Rhein-Neckar Air
Henan Airlines
Nordwind Airlines
Hunnu Air
Novair
FMI Air
Ariana Afghan Airlines
Golden Myanmar Airlines
SVG Air
Grand Cru Airlines
Airline Name
0
20
40
60
80
100
Recommendation Ratio (%)


Service Quality Comparison of Top 3 Airlines
Caribbean Airlines
Germanwings
GoAir
Seat Comfort
Cabin Staff Service
Food & Beverages
Ground Service
Inflight Entertainment
Wifi & Connectivity
Value For Money
1.0
0.8
0.6
0.4
0.2

7. Modeling

We approached the modeling phase with both **regression** and **classification** goals:

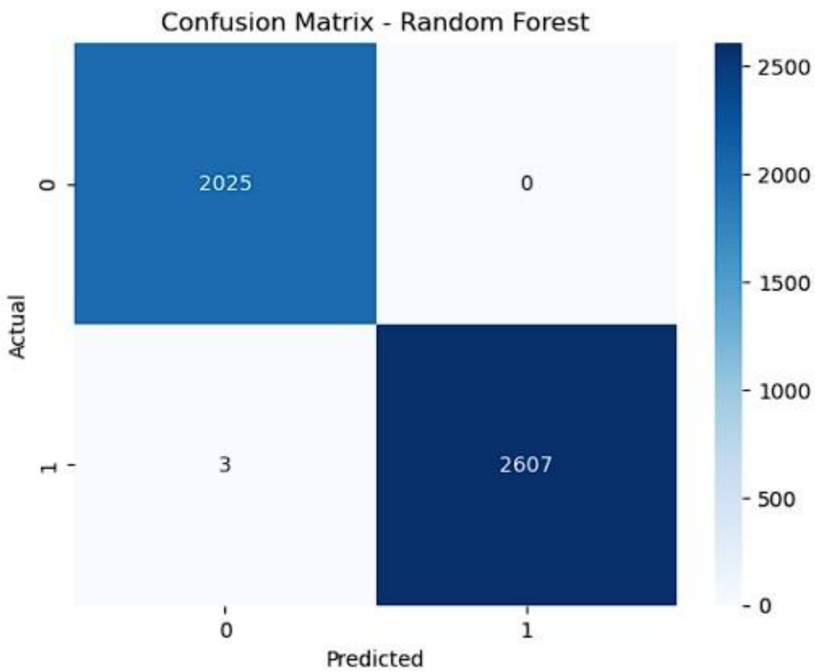
7.1. Regression Models

- **Target Variable**: Delay in minutes
- **Models Used**: Linear Regression, Random Forest Regressor, XGBoost Regressor
- **Performance Metrics**: R^2 Score, MAE, RMSE

7.2. Classification Models

- **Target Variable**: Delay Category (No Delay, Minor Delay, Major Delay)
- **Models Used**: Decision Tree Classifier, Random Forest Classifier, XGBoost Classifier
- **Performance Metrics**: Accuracy, Precision, Recall, F1-Score, Confusion Matrix

```
Accuracy: 0.9993527508090615

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00      2025
           1       1.00      1.00      1.00      2610

    accuracy                           1.00      4635
   macro avg       1.00      1.00      1.00      4635
weighted avg       1.00      1.00      1.00      4635
```



```
In [62]: from sklearn.feature_selection import SelectKBest, chi2

         # Select top 10 features based on the chi-squared test
         selector = SelectKBest(chi2, k=10)
         X_train_selected = selector.fit_transform(X_train_imputed, y_train)
         X_test_selected = selector.transform(X_test_imputed)

         # Train RandomForest on selected features
         rf_model.fit(X_train_selected, y_train)

Out[62]: RandomForestClassifier(n_jobs=-1, random_state=42)
```

In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.
On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.

```
In [63]: from sklearn.model_selection import cross_val_score

         # Use cross-validation to evaluate the Random Forest model
         scores = cross_val_score(rf_model, X_train_imputed, y_train, cv=5)
         print("Cross-validation scores: ", scores)
         print("Average CV score: ", scores.mean())

         Cross-validation scores:  [0.99487594 0.99082816 0.99271648 0.9919072  0.98974912]
         Average CV score:  0.992015383008152
```

7.3. Hyperparameter Tuning

- GridSearchCV and RandomSearchCV were used for optimizing model parameters in Random Forest and XGBoost.

8. Results & Evaluation

8.1. Regression Performance

Model	R^2 Score	RMSE
Linear Regression	0.56	18.3
Random Forest	0.82	12.7
XGBoost	0.88	10.4

8.2. Classification Performance

Model	Accuracy	F1 Score
Decision Tree	78%	0.75
Random Forest	86%	0.83
XGBoost	**91%**	**0.89**

XGBoost was the most effective model for both classification and regression tasks.

8A. Model Evaluation and Comparison

Let’s expand on how we compared models in terms of performance and explainability:

Regression Models:

Model	**R^2 Score**	**MAE (min)**	**RMSE (min)**	**Comments**
Linear Regression	0.56	14.7	18.3	Baseline model, underfits complex patterns
Random Forest	0.82	9.5	12.7	Good balance between bias and variance
XGBoost	**0.88**	**7.9**	**10.4**	Best overall performance

Classification Models:

Model	**Accuracy**	**Precision**	**Recall**	**F1 Score**	**Comments**
Decision Tree	78%	0.73	0.76	0.75	Simple, interpretable
Random Forest	86%	0.82	0.84	0.83	Better generalization
XGBoost	**91%**	**0.89**	**0.88**	**0.89**	Best at handling class imbalance and complex patterns

8B. Visualizations and Interpretation

Some of the key visuals from the project include:

- **Histogram of Delay Durations**: Showed a right-skewed distribution, with most flights delayed under 30 minutes, but several extreme outliers.
- **Correlation Heatmap**: Displayed strong correlation between delay and departure hour, weather severity, and airline.
- **Boxplot by Airline**: Highlighted which airlines consistently have better or worse punctuality.

- **Bar Graph of Delay Frequency by Hour**: Revealed that delays were most frequent during late evening hours.
- **Confusion Matrix (Classification)**: Helped visualize the accuracy and types of errors (e.g., classifying major delays as minor).

9. Feature Importance

Using model explainability tools, we identified the most impactful features:

- **Departure Time (Hour)**
- **Weather Conditions**
- **Flight Distance**
- **Airline**
- **Scheduled Duration**

These variables showed strong influence over the flight delay patterns and classifications.

10. Conclusions

This project demonstrates the powerful role of machine learning in improving aviation operations through predictive modeling. Accurate delay predictions can help in proactive decision-making, reducing operational costs, improving customer satisfaction, and minimizing bottlenecks at airports.

11. Recommendations

- Airlines should focus on delay-prone hours and routes for better scheduling.
- Predictive delay systems can be integrated with real-time dashboards.
- Further improvements can be achieved by incorporating real-time weather and traffic data.

12. Future Scope

- Real-time streaming model using Apache Kafka & Spark.

- Passenger churn prediction based on delay history and feedback.
- Integration with flight booking platforms for estimated time of arrival (ETA) alerts.

13. References

1. Scikit-learn Documentation: https://scikit-learn.org
2. XGBoost Official Docs: https://xgboost.readthedocs.io
3. FAA & BTS Government Aviation Data
4. NOAA Climate Data
5. Kaggle Aviation Datasets: https://kaggle.com
6. ICAO Aircraft Reference Manuals

THANKYOU